

YASHASWI VEMU

Software Engineer

SUMMARY

Software Engineer with 2+ years owning scalable distributed systems using Java, Spring Boot, and AWS. Built AI-powered automation at ServiceNow across 500+ enterprise deployments and event-driven fintech platforms at Paves focused on customer impact, operational excellence, and systems that scale.

PROFESSIONAL EXPERIENCE

Software Engineer, Paves Technologies – Dallas, TX | Sep 2025 – Present

- Designed and maintained REST APIs and distributed backend services using Java, Python, and GCP for payment processing and internal transaction workflows
- Applied Machine Learning models using Python to strengthen transaction monitoring and generate prediction scores for suspicious payment activity.
- Built Ranking workflows using Machine Learning predictions to prioritize flagged transactions for fraud and operations review.
- Built internal React UI components for fraud review and transaction validation workflows, connecting them with backend APIs for faster case handling.
- Integrated Generative AI and Natural Language Processing (NLP) using Vertex AI to extract key fields from compliance and verification documents.
- Exposed Machine Learning model outputs through backend services, enabling prediction results in internal APIs and monitoring tools.
- Built internal data pipelines for verification data and supported Speech and Audio processing flows using NLP models for identity checks.
- Developed GCP service integrations using Java and Python to support payment workflows, validation checks, and compliance processing.
- Supported the design, testing, deployment, and maintenance of Java and Python services on GCP, helping improve reliability across high-volume payment workflows.

Software Engineer, ServiceNow – Hyderabad, India | Jul 2022 – Jul 2024

- Contributed to the Merge Agent, an AI agent that automates metadata conflict resolution during ServiceNow platform upgrades, reducing manual effort in the upgrade pipeline.
- Built a custom Fluent SDK plugin, scan check skills, and 3-way merge skills to help the Merge Agent understand metadata, detect conflicts early, and reconcile upgrade changes.
- Integrated the Merge Agent with an MCP server and internal build tools, enabling automated fetch and update of upgrade records in the build pipeline.
- Worked on Advanced Search for Application Manager using React and JavaScript, improving Information Retrieval and speeding up component discovery by 25%.
- Designed ServiceNow platform features, Upgrade Console components, and backend integrations using JavaScript and REST APIs to improve upgrade workflow reliability.
- Developed UI components using ServiceNow UI Builder, JavaScript, and React, built Advanced Search for Application Manager.
- Authored, tested, and deployed skills in ServiceNow AI Agent Studio as part of a production agent workflow.
- Implemented automated frontend and backend tests, reducing regression issues during platform upgrades.

PROJECTS

AthletaAI – Distributed E-commerce Platform

UTA | Fall 2025

Technologies: Java, Spring Boot, React, AWS SQS, DynamoDB, Docker

- Designed a distributed e-commerce platform using **Java Spring Boot microservices** and React for product browsing, auth, and order processing — designed for horizontal scalability and fault isolation.
- Implemented event-driven order workflows with **AWS SQS** for async order/payment/inventory communication; RESTful APIs backed by **DynamoDB** and containerized with Docker; achieved 99.9% availability under load.

Visual Question Answering System

UTA | Spring 2025

Technologies: Python, TensorFlow, CNN, LSTM, NLP

- Developed a multimodal **deep learning system** combining CNN (vision) and LSTM (NLP) to answer image-based questions; applied ML optimization techniques to improve prediction accuracy.

CONTACT

 Dallas, TX
 (214) 709-0938
 vemuyash@gmail.com
 [LinkedIn](#)
 [Portfolio](#)

EDUCATION

Master of Science

Computer Science

University of Texas at Arlington

Arlington, TX | May 2026

Relevant Coursework:

Data Structures & Algorithms, Distributed Systems, Machine Learning, OS, Database Systems, Software Engineering

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, C, C++, SQL

Frameworks: React.js, Spring Boot, Node.js, Express.js, Next.js

Cloud : Google Cloud Platform (GCP), Vertex AI, Pub/Sub, BigQuery, AWS (SQS, DynamoDB, EC2, S3, Lambda, CloudFormation)

DevOps: Docker, Git, CI/CD, Jenkins, Bash, Linux

Databases: MySQL, MongoDB, DynamoDB, PostgreSQL

Architecture: REST APIs, Microservices, WebSockets, Event-Driven

AI & Agents: Generative AI, ServiceNow AI Agent Studio, MCP, Fluent SDK, Machine Learning, Machine Learning Algorithms, Natural Language Processing (NLP), Recommendations, Ranking, Predictions, Machine.

Practices: Agile/Scrum, Test Automation, Distributed Systems.

EVIDENCE OF EXCELLENCE

- Built an AI-powered Merge Agent on ServiceNow's enterprise platform, reducing manual developer effort by ~40% across 500+ customer deployments.
- Integrated Generative AI workflows at Paves Technologies to automate compliance checks, reducing manual review and improving processing efficiency.
- Developed a distributed Java Spring Boot e-commerce platform with SQS, DynamoDB, and Docker, supporting 99.9% availability under high load.